

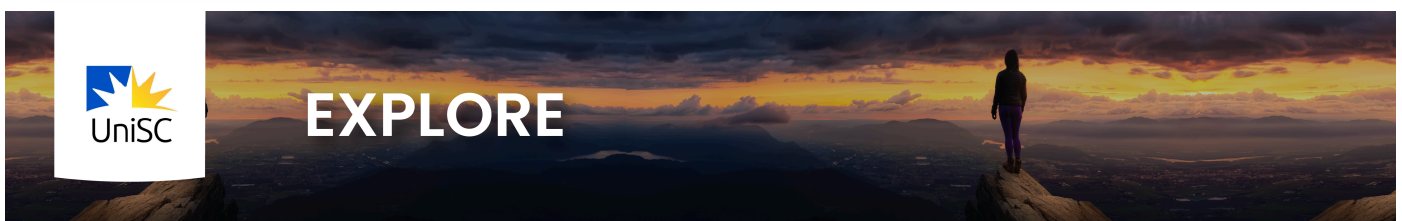
3.1 Measuring Portfolio Returns



Measuring Portfolio Returns

Guiding Questions

1. What information do we need to calculate the Holding Period Return (HPR) for a portfolio?
2. How does HPR incorporate both capital gains and income (dividends or interest)?
3. What does HPR tell us about portfolio performance over a single investment period?
4. Why can a simple average of returns give a misleading picture of performance across multiple years?
5. How does the geometric mean capture the effect of compounding in long-term investment growth?
6. Why is measuring return alone insufficient for evaluating portfolio performance without considering risk or inflation?



What will you learn?

You will be introduced to measuring portfolio return. You'll learn:

- How to calculate the Holding Period Return (HPR) for a portfolio.
 - How portfolio returns combine capital gains and income (dividends or interest).
 - How to interpret returns over a single investment period.
 - The difference between arithmetic and geometric average returns.
 - Why compounding matters when evaluating multi-period investment performance.
 - Why return measurement is the starting point for portfolio evaluation, before considering risk or inflation.
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3.1 Measuring Portfolio Returns

1. Why Do We Measure Portfolio Returns?

Portfolio return represents the financial reward earned from investing capital over a given period. Measuring returns is the first step in evaluating portfolio performance, because it tells us how much value the investment generated.

However, measuring return is not always straightforward. Investors may want to evaluate performance over a single period or over multiple periods, and the appropriate measure differs depending on the purpose of the analysis.

Key reasons for measuring portfolio returns include:

- Assessing investment performance
- Comparing results with benchmarks or other portfolios
- Understanding how investments contribute to wealth accumulation

Before considering risk or inflation, we must first determine how much the portfolio actually earned.

2. Holding Period Return (HPR)

What is Holding Period Return?

The Holding Period Return (HPR) measures the total percentage return on an investment over a single time period.

It includes two components:

1. Capital gain (or loss): change in the asset's price
2. Income received: dividends, interest, or coupons

Thus, HPR reflects the total payoff received by the investor during the holding period.

HPR Formula

$$HPR = \frac{(P_1 - P_0) + D}{P_0}$$

Where:

- P_0 = Initial portfolio value
 - P_1 = Ending portfolio value
 - D = Income received (dividends or interest)
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Example 1: Calculating Holding Period Return

An investor buys a portfolio worth \$30,000.

After one year:

- Portfolio value = \$32,400
- Dividends received = \$600

Step 1: Calculate total gain

Capital gain: $32,400 - 30,000 = 2,400$

Total gain: $2,400 + 600 = 3,000$

Step 2: Divide by initial investment

$HPR = 3,000 / 30,000 = 0.10$

Result: $HPR = 10\%$

This means the portfolio generated a 10% total return during the year.

Interpretation

HPR answers the question:

“What happened to the investment during this period?”

However, HPR alone does not tell us:

- Whether the return was good or bad
- Whether the return compensated for risk
- Whether the return exceeded inflation
- Whether the portfolio outperformed the market

These questions will be addressed in later sections.

2.1 Portfolio Return

In practice, portfolios usually contain multiple assets rather than a single investment. Therefore, we need a method for calculating the overall return of the portfolio based on the returns of individual assets.

Portfolio return is calculated as the weighted average of the returns of the assets held in the portfolio.

Assets that represent a larger share of the portfolio have a greater impact on the total return.

Portfolio Return Formula

$$R_p = \sum_{i=1}^n w_i R_i$$

Where:

R_p = portfolio return

w_i = weight of asset i in the portfolio

R_i = return of asset i

The weights represent the proportion of the portfolio invested in each asset, and they must sum to one.

$$\sum_i w_i = 1$$

Example: Portfolio Weighted Return

Suppose an investor allocates funds as follows:

Asset	Weight	Return
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Stock A	40%	12%
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Stock B	35%	8%
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Asset Weight Return

ETF C 25% 10%

Step 1: Multiply each return by its weight

Stock A contribution

$$0.40 \times 12\% = 4.8\%$$

Stock B contribution

$$0.35 \times 8\% = 2.8\%$$

ETF C contribution

$$0.25 \times 10\% = 2.5\%$$

Step 2: Add contributions

$$R_p = 4.8 + 2.8 + 2.5 = 10.1\%$$

Interpretation

The portfolio generated a 10.1% return, meaning that an investment of \$100 would grow to approximately \$110.10 over the period.

This measure captures the combined effect of all investments in the portfolio.

3. Multi-Period Returns

Many investments are held for several years, so we need a way to summarize performance across multiple periods.

A common mistake is to simply calculate the arithmetic average return.

However, this can be misleading because investment returns compound over time.

4. Arithmetic vs Geometric Returns

Arithmetic Average Return

The arithmetic average is calculated as:

$$\text{Average Return} = \frac{R_1 + R_2 + \dots + R_n}{n}$$

This method assumes returns are independent and additive, which ignores compounding.

It is useful for short-term expectations, but not for measuring actual wealth growth.

Geometric Mean Return

The **geometric mean** measures the **true compound growth rate** of an investment.

Formula:

$$(1 + R_1)(1 + R_2) \cdots (1 + R_n)^{1/n} - 1$$

This provides the **constant annual return that would produce the same final value**.

Example 2: Arithmetic vs Geometric Returns

Suppose a portfolio produces the following returns:

Year 1: **+12%**

Year 2: **-5%**

Year 3: **+8%**

Arithmetic Average

$$\frac{12-5+8}{3} = 5\%$$

Geometric Mean

Convert to decimals:

$$(1.12)(0.95)(1.08)$$

Multiply:

$$1.148$$

Take cube root:

$$1.148^{1/3} = 1.0474$$

Subtract 1:

$$0.0474$$

Result

Geometric return $\approx$ **4.74%**

Interpretation

- Arithmetic average = **5%**
- Geometric average = **4.74%**

The arithmetic average **overstates the true growth rate** because it ignores compounding and the effect of losses.

This difference occurs because:

Losses reduce the investment base, making recovery more difficult.

For this reason:

- **Geometric returns are preferred for long-term performance evaluation.**
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5. Key Insights from Return Measurement

From this section we learn:

1. Portfolio return measures **how much an investment earned**.
2. **Holding Period Return (HPR)** is used for single-period analysis.
3. Multi-period performance requires **compounding**.
4. **Arithmetic averages can overstate performance**.
5. **Geometric mean returns provide the true growth rate of wealth**.

Return measurement therefore forms the **foundation of portfolio evaluation**, but it must later be combined with **risk and inflation analysis**.

Reflect

1. What two components make up the total return of an investment?
2. What does the Holding Period Return measure?
3. Why can the arithmetic average return be misleading for long-term investment evaluation?
4. Why is the geometric mean a better measure of multi-period performance?
5. What important aspects of portfolio performance are **not captured** by simple return measures?

Solve

Question 1

A portfolio is purchased for **\$50,000**.

After one year:

- Portfolio value = **\$52,000**
- Dividends received = **\$2,000**

Calculate the **Holding Period Return**.

Question 2

A portfolio is initially worth **\$500,000**.

At the end of the year:

- Value = **\$545,000**
- Dividends = **\$15,000**

Calculate the **Holding Period Return**.

Question 3

A portfolio earns the following annual returns:

Year 1: **20%**

Year 2: **-10%**

Calculate:

- a) Arithmetic average return
 - b) Geometric average return
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Question 4

Explain why a 10% loss requires more than a 10% gain to recover.

Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3292583?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292583?wrap=1).

Critically evaluate

Two portfolios both report an **average annual return of 8%**.

Portfolio A experienced **large fluctuations**, while Portfolio B produced **stable returns**.

Which portfolio likely has the **higher geometric return**, and why?